Mahdi Salem,

Lead Power BI Developer | Data Architect

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Architecting enterprise analytics — from fragmented operational data to governed, decision-ready intelligence.

Professional Profile

Lead Power BI Developer, Data Architect, and Senior Data Engineer with over 10 years of experience delivering enterprise analytics, operational reporting, data integration, and process automation — including eight years supporting complex aerospace manufacturing operations at Bombardier. Holds Microsoft certifications in Fabric Analytics Engineering (DP-600), Power BI Data Analysis (PL-300), and Fabric Data Engineering (DP-700), with deep hands-on expertise designing Direct Lake semantic models, Star Schema dimensional architectures, and enterprise Power BI governance frameworks across Microsoft Fabric, Databricks, and Azure Synapse. Combines rigorous data modeling and engineering depth with business analysis and cross-functional team leadership.

Proven track record of architecting trusted reporting foundations from fragmented operational sources, establishing enterprise data governance, and transforming complex manufacturing metrics into actionable executive insights. Experienced in designing robust data pipelines (SSIS, Fabric Data Factory, Dataflow Gen2) and high-performance semantic models across multi-factory environments. At Bombardier, established the foundation for digitized reporting by leading the cross-functional data migration from legacy on-premises systems (including Oracle 10g and SQL Server) to modern cloud architectures across Azure Synapse, Databricks, and Microsoft Fabric — delivering a unified, governed analytical layer that significantly boosted reporting efficiency and reduced departmental data friction.

Highlights of Professional Qualifications

- **Delivered 150+ Power BI reports, dashboards, and PowerApps solutions** for Production Andon, Quality Assurance, and PMO governance — establishing standardized semantic models, reusable design patterns, and enterprise metric consistency.
- **Architected full-lifecycle BI & analytical solutions:** stakeholder discovery, Star Schema dimensional modeling, Data Vault 2.0 principles, DAX measure optimization, Direct Lake serving layers, RLS/OLS security, Premium capacity governance, and CI/CD deployment.
- **Microsoft Certified across Fabric, Power BI, Azure, and Power Platform:** Fabric Analytics Engineer (DP-600), Power BI Data Analyst (PL-300), Fabric Data Engineer (DP-700), Azure Data (DP-900), and App Maker (PL-100); experienced architecting scalable solutions on Microsoft Fabric, Databricks, and Power Platform.
- **10+ years of consultative business analysis and data strategy:** partnering with Executive, Operations, Finance ... stakeholders to translate business goals into governed data flows, logical/physical models, and executive analytics.
- **Led and mentored cross-functional teams** of up to 4 developers and data engineers, instituting development standards, code reviews, role clarity, and structured technical capability building.
- **Defined architecture specifications and resolved complex data reconciliation issues** across Enterprise Data Warehouse, Master Data Management (MDM), and operational systems (SAP S/4HANA, MES, legacy DBs), ensuring strict data integrity and SLA adherence.
- **Advanced analytical tooling and semantic model optimization:** expertise in DAX, Power Query (M), Direct Lake performance tuning, incremental refresh, XMLA endpoint management, and automated monitoring (Python, PowerShell).

Professional Experience

Lead Business Intelligence Architect & Data Engineer **Aug 2019 – Sep 2026**

Bombardier Aerospace *Mississauga, ON*

- Engineered and deployed over 150 Power BI reports, dashboards, and PowerApps solutions supporting Production Andon teams, Quality Assurance, and PMO governance — establishing standardized semantic models, reusable report templates, and UX mock-up frameworks under the Bombardier Operational Excellence (BOE) initiative.
- Architected the enterprise analytics platform modernization into Microsoft Fabric: designed Direct Lake–enabled semantic models, onboarding 15+ data sources via Data Pipeline and 20+ via Dataflow Gen2 into OneLake to deliver curated datamarts and self-service reporting layers.
- Led data platform ingestion and transformation of 20+ operational sources (SAP S/4HANA, MES, legacy DBs) into Azure Synapse Analytics and Databricks, orchestrating medallion-layer data refinement for downstream analytical reporting.
- Initiated and implemented a Data Quality KPI framework reconciling SAP, MES, and local production databases, materially improving shop-floor data integrity and establishing shared metric definitions across business units.
- Integrated Azure Data Lake and Azure Portal services to monitor and optimize Power BI Premium capacity utilization, reducing resource overhead and query latency while safeguarding concurrent user performance.
- Defined and enforced data governance and security frameworks, including Row-Level and Object-Level Security (RLS/OLS), data masking, and metadata catalogs across enterprise reporting datasets.
- Designed, developed, and maintained enterprise-grade SSIS packages across a multi-factory, multi-country footprint, optimizing data-flow efficiency and eliminating operational pipeline bottlenecks.
- Designed and launched an adaptive OSW (Outstanding Work) Production Tracking application (.NET / Angular) with an integrated attendance management dashboard (Python, Django, SQL) to structure highly volatile shop-floor assembly tasks.
- Led a team of up to 4 developers and data engineers through detailed sprint planning and mentoring; established reusable development standards and knowledge transfer sessions to build independent team capability.
- Automated operational workflows through 10+ PowerApps solutions and scripted tooling, eliminating manual administrative entry and accelerating cross-departmental coordination.

Senior Data and Process Analyst **Jul 2018 – Aug 2019**

Triumph Aerospace Group *Toronto, ON*

- Architected a multi-tiered reporting framework covering daily, weekly, and monthly operational metrics at macro and aircraft-structural levels, providing actionable visibility to executive and shop-floor leaders.
- Engineered automated Crew Load Chart generation and reporting workflows, reducing compilation turnaround time from 3 hours to 2 minutes and eliminating manual data-entry errors.
- Built a Python-based web platform for daily attendance management with integrated Crew Load analytics to measure shift efficiency and optimize labor allocation.
- *Developed real-time alerting mechanisms and KPI dashboards delivering threshold notifications to operations managers to proactively prevent assembly delays.*

Process Improvement Lead **Aug 2016 – Jun 2018**

AAA Canada (Aerospace & Transportation) *Toronto, ON*

- Built a Manpower Mapping application distinguishing touch vs. non-touch labor, improving shop-floor labor

transparency and reducing non-value-added idle time.

Architected a Line Logistic Cage tracking system providing real-time visibility on job kits and workability post-punch-out, preventing downstream assembly bottlenecks.

Developed an automated Rapid Non-Conformance (RNC) reporting tool that shortened generation time to 3 minutes, accelerating quality-containment workflows.

Digitized Daily Management Meeting (DMM) reporting into an automated dashboard, increasing reporting efficiency by 200% while eliminating manual data preparation.

Lead Business Consultant **Mar 2001 – Jun 2015**

Various Companies (Independent) *Tehran, Iran*

Directed 30+ consulting engagements in strategic planning, business process re-engineering, TQM and Six Sigma, performance measurement, and organizational improvement across multiple industries.

Prepared more than 100 proposals and RFQs; supported clients with performance measurement and organizational improvement.

Technical Expertise

BI & Analytics	Power BI (Direct Lake, Copilot, Semantic Models, Dataflows Gen2, RLS/OLS, Incremental Refresh, XMLA, Paginated Reports), Dataiku, DAX, Power Query (M), Tableau, SAS, Cognos
Data Engineering & Architecture	Microsoft Fabric (OneLake, Lakehouse, Warehouse, Data Factory, Dataflow Gen2), Databricks (Delta Lake, Unity Catalog, PySpark), Azure Synapse Analytics, SSIS, Kafka, dbt, ETL/ELT, Medallion Architecture, Kimball Dimensional Modeling (Star/Snowflake), Data Vault 2.0, SCD Type 1/2/3/6
Data Governance & DevOps	Microsoft Purview, Data Governance & Lineage, Data Masking (DDM), Data Classification, Git, GitHub, Azure DevOps Pipelines, CI/CD for BI, PowerShell, Bash
Databases & Infrastructure	SQL Server (T-SQL, SSMS, SSIS, SSAS, SSRS, RLS/OLS), PostgreSQL, MySQL, Oracle (10g/12c, VPD), Azure Cosmos DB, Docker, Linux/Unix, REST APIs / JSON
Programming & Data Science	Python (PySpark, Pandas, NumPy, SciPy, scikit-learn, Django, Flask), T-SQL, PL/SQL, KQL, R, Jinja2
Application Development & Automation	PowerApps, Dynamics 365, Power Automate, Power BI Embedded, React, Angular, TypeScript, ASP.NET Core, Node.js/Express, HTML5/CSS3

Education & Training

Master of Business Management, Marketing

Allameh Tabatabaie University, 1999 – 2002

Bachelor's Degree, Business Management

Allameh Tabatabaie University, 1996 – 1999

Professional Training

- Business Analytics; Duke University, 2016 – 2017
- CS50's Web Programming with Python and JavaScript; Harvard University, 2020
- Microsoft Power Platform (Power BI, PowerApps, Power Automate); 2020 – 2021

Certifications

Microsoft Certified:

- Power BI Data Analyst Associate (PL-300)
- Fabric Analytics Engineer Associate (DP-600)
- Data Engineering in Fabric Associate (DP-700)
- Power Platform App Maker Associate (PL-100)
- Power Platform Fundamentals (PL-900)
- Azure Data Fundamentals (DP-900)
- Excel 2016 Expert

- Certified ScrumMaster (CSM I)

Six Microsoft badges held in total; some pending renewal.